DSP-M.T. Exam – Model -A – SOL- ٤ / ١١ / ٢٠١٩

Q # ١- SOL.- Cont.-٢

c- Given the analog signal: $x(t) = 5\cos(2000\pi t)$ to be sampled with sampling frequency F_S = 3000Hz.

(i) Using the sampling frequency given above, could the original signal be reconstructed successfully from its samples?

(ii) Take F_S = 6000Hz, find an expression for its corresponding sequence, x(n). Then calculate its fundamental period N_P.

(iii) Find the samples values of the first cycle in x(n), obtained part (i), then plot *x(t)* and x(n) on the same figure.

SOLUTION

(i) $x(t) = 5\cos(2000\pi t)$

$x(t) = A\cos(2\pi Ft + \theta)$ ∴ $A = 5$ V , $\theta = 0$ rad $2\pi Ft = 2000\pi t$ $F = 1000$ HZ

$F_{max} = 1000$ Hz $F_S \geq 2F_{max}$

$F_S \geq 2 \cdot 1000$ $F_S \geq 2000$ HZ

we use $F_S = 3000$ HZ

Yes, we can Reconstructed successfully the original signal From its Sample

(ii) at $F_S = 6000$ HZ

$t \longrightarrow nT_S = \frac{n}{F_S}$ $F_S = 6000$ $t \longrightarrow \frac{n}{6000}$

$x_a(nT_S) = 5\cos\left(\frac{2000\pi \cdot n}{6000}\right)$

$$\boxed{x(n) = 5\cos\left(\frac{n\pi}{3}\right)}$$

To get $N_P \rightarrow x(n) = A\cos(2\pi Fn)$ the Fundamental Period N_P

$2\pi Fn = \frac{n\pi}{3}$ $\therefore F = \frac{1}{6} = \frac{K}{N_P}$ $\boxed{\therefore N_P = 6}$ samples /cycle.

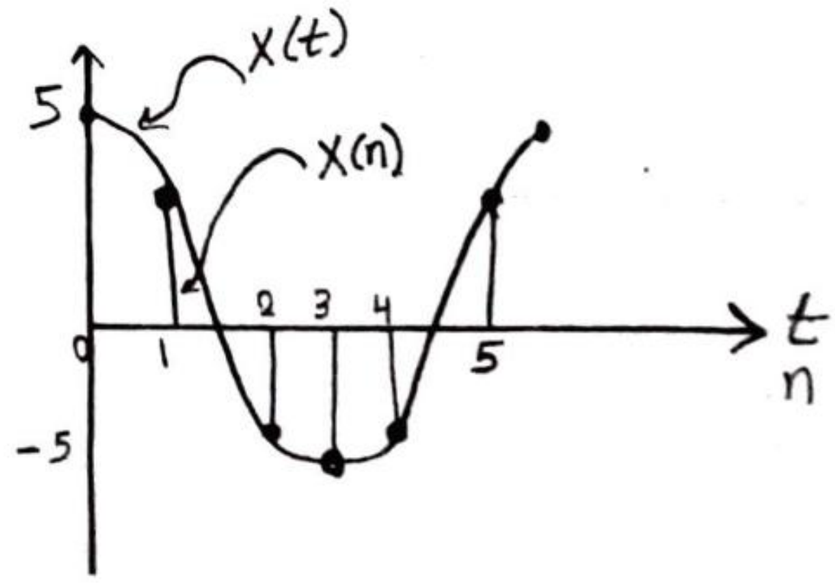
(iii) $x(n) = 5\cos\left(\frac{n\pi}{3}\right)$

$N_P = 6$ $n = 0 \ldots N_P - 1 = 0, 1, 2, 3, 4, 5$

$x(0) = 5\cos(0) = 5$ { $x(3) = 5\cos(\pi) = -5$

$x(1) = 5\cos\left(\frac{\pi}{3}\right) = 2.5$ { $x(4) = 5\cos\left(\frac{4\pi}{3}\right) = -2.5$

$x(2) = 5\cos\left(\frac{2\pi}{3}\right) = -2.5$ { $x(5) = 5\cos\left(\frac{5\pi}{3}\right) = 2.5$



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